Knowledge Reasoning & Knowledge Graph Embedding — Report

Lab Session 5 — Cinema Domain Knowledge Base

Web Mining & Semantics

March 25, 2026

Part 1 — Knowledge Reasoning with SWRL

Ontology and Rule

The `family.owl` ontology defines a family domain with classes including `Person`, `Male`, `Female`, `Parent`, `Child`, and object properties such as `isFatherOf`, `isMotherOf`, `isSiblingOf`, and the datatype property `age : Person → xsd:int`. The ontology was loaded using OWLReady2 and extended with a new class `oldPerson` as a subclass of `Person`.

The following SWRL rule was defined and applied:

$$\texttt{Person}(?p) \wedge \texttt{age}(?p, ?age) \wedge \texttt{greaterThan}(?age, 60) \rightarrow \texttt{oldPerson}(?p)$$

Reasoning Result

We got this table

Table 1: SWRL inference results — `oldPerson` class

Individual	Age	Inferred as `oldPerson`
Peter	70	Yes
Marie	69	Yes
John	45	No
Thomas	40	No
Sylvie	30	No
Paul	38	No
Alex	25	No
Chloé	18	No
Tom	10	No
Pedro	10	No
Michael	5	No
Claude	5	No

The rule correctly identifies Peter (70) and Marie (69) as `oldPerson` instances while excluding all individuals aged 60 or below.

Part 2 — Knowledge Graph Embedding

Data Preparation

The expanded KB (`expanded_kb.nt`, 99,106 triples) from the previous lab was filtered to retain only the 7 cinema-relevant Wikidata predicates (Table 2), removing all metadata and literal-valued triples. The filtering produced a compact but clean dataset of **276 triples**, **250 unique entities**, and **6 unique relations**. The reduced triple count reflects a key challenge discussed in Section 3: most of the 99,106 raw triples consisted of Wikidata metadata predicates unrelated to the cinema domain.

Table 2: Retained Wikidata predicates

Property	Label	Private predicate
P57	director	`:direct`
P161	cast member	`:star`
P162	producer	`:produce`
P166	award received	`:earn`
P364	original language	—
P495	country of origin	—
P577	publication date	—

Train / Validation / Test Split

The 276 triples were randomly shuffled (seed=42) and split as follows: **220 train**, **3 validation**, **8 test**. All validation and test entities were verified to appear in the training set to prevent information leakage.

Training Configuration

Both models were trained with identical hyperparameters for a fair comparison:

- Embedding dimension: 200
- Learning rate: 0.01
- Batch size: 512
- Epochs: 500
- Negative sampling: default PyKEEN strategy
- Device: CUDA GPU (Google Colab T4)
- Framework: PyKEEN

Evaluation Results — Link Prediction

Table 3 reports filtered metrics on the test set for both head and tail prediction averaged ("both — realistic").

Table 3: Link prediction results (filtered metrics, test set)

Model	MRR	Hits@1	Hits@3	Hits@10
TransE	0.1683	0.1250	0.1250	0.1875
ComplEx	0.0156	0.0000	0.0000	0.0000

TransE outperforms ComplEx across all metrics. This is a surprising result, since ComplEx is theoretically better suited for the asymmetric relations in the cinema domain. The reason is likely the extremely small dataset size (220 training triples): ComplEx has 2× more parameters per entity than TransE (complex vs. real vectors) and requires significantly more data to converge. TransE, with its simpler translation objective $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$, generalises better on sparse datasets.

KB Size Sensitivity

Table 4 shows MRR for TransE trained on subsets of the training data.

All three conditions produce identical MRR (0.1956). This is expected: the training set contains only 220 triples, so the "20k" and "50k" subsets both reduce to the full 220-triple set. This result confirms that the filtered KB is the binding constraint, not the subsampling. A larger, less aggressively filtered KB would be required to observe meaningful size sensitivity.

Table 4: TransE MRR vs. training set size

Subset	MRR
20k triples	0.1956
50k triples	0.1956
Full (220)	0.1956

Embedding Analysis

Nearest neighbour analysis and clustering were performed using the **TransE** model, which achieved superior link prediction performance (MRR = 0.1683) compared to ComplEx (MRR = 0.0156) on this dataset. TransE embeddings are real-valued (dim = 200), and similarity was computed as cosine similarity over the full embedding vectors.

Nearest Neighbors

Table 5 reports the 5 nearest neighbours for each core cinema entity in the TransE embedding space.

The results demonstrate strong semantic coherence across all queried entities, with cosine similarities ranging from 0.24 to 0.59 — a substantial improvement over the ComplEx results (0.10–0.19). Several particularly meaningful patterns emerge:

- **Christopher Nolan** neighbours include Inception (his film), Michael Caine (his regular collaborator across 6 films), Leo DiCaprio (Inception lead), and Stanley Kubrick (fellow auteur director).
- **The Godfather** neighbours include Marlon Brando, Al Martino, and Johnny Martino — all cast members — as well as the Golden Globe Award for Best Screenplay and Best Picture, both awards the film received.
- **Stanley Kubrick** neighbours include Francis Ford Coppola and Christopher Nolan, two directors of comparable stature, as well as Gary Lockwood and Keir Dullea, the lead actors of *2001: A Space Odyssey*.
- **Martin Scorsese** neighbours are dominated by prestigious lifetime achievement awards (Kennedy Center Honors, BAFTA Fellowship, Legion of Honour, Library of Congress Living Legend), reflecting that the KB expansion captured award-related triples for this entity.
- **Pulp Fiction** and **Inception** neighbours are predominantly awards and cast members, consistent with the P166 (award received) and P161 (cast member) predicates dominating the filtered KB.

These results confirm that TransE has successfully learned relational structure from the cinema domain despite training on only 220 triples, capturing both person-film co-occurrence and award-entity associations.

Clustering Analysis

A t-SNE projection (2D, perplexity=30) was computed on the 250 entity embeddings (TransE, real-valued, dim=200). Due to the very small entity count and the dominance of two relation types (award received and cast member) in the filtered triples, no strong class-based clustering was observed. Directors and films did not form clearly separated clusters, suggesting the embedding space is primarily organised around relational proximity rather than ontological class membership. This is a known limitation of translational models on sparse, relation-homogeneous

Table 5: Nearest neighbours in TransE embedding space (cosine similarity)

Query entity	**Neighbour**	**Sim.**
Christopher Nolan	Inception	0.416
	Michael Caine	0.374
	Leo DiCaprio	0.371
	Stanley Kubrick	0.353
	Taylor Geare	0.336
The Godfather	Al Martino	0.393
	Marlon Brando	0.379
	Golden Globe Award for Best Screenplay	0.353
	Johnny Martino	0.353
	Best Picture	0.333
Stanley Kubrick	Francis Ford Coppola	0.439
	Christopher Nolan	0.353
	Bill Weston	0.314
	Gary Lockwood	0.253
	Keir Dullea	0.240
Martin Scorsese	Library of Congress Living Legend	0.588
	Kennedy Center Honors	0.493
	BAFTA Fellowship	0.454
	Officer of the Legion of Honour	0.441
	David di Donatello Lifetime Achievement	0.440
Pulp Fiction	David di Donatello for Best Foreign Actor	0.541
	BAFTA Award for Best Supporting Actor	0.523
	Independent Spirit Award Best Screenplay	0.459
	Maria de Medeiros	0.453
	Julia Sweeney	0.401
Inception	Yuji Okumoto	0.543
	Academy Award for Best Sound Editing	0.517
	Leo DiCaprio	0.499
	Best Cinematography	0.496
	Michael Gaston	0.487

KBs: without diverse relation types connecting entities of the same class, the model has no signal to group them geometrically.

Clustering Analysis

A t-SNE projection (2D, perplexity=30) was computed on the 250 entity embeddings (TransE, real-valued, dim=200). Figure 1 shows the resulting scatter plot.

No strong class-based clustering is observable. Directors and films do not form clearly separated clusters, suggesting the embedding space is organised around relational proximity rather than ontological class membership.

Relation Behaviour

The 6 retained relations are all **asymmetric**: director, cast member, producer, award received, original language, and country of origin all have a clear directional interpretation (film $\rightarrow$ person or film $\rightarrow$ attribute). TransE models these as translation vectors $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$ and handles asymmetry through vector directionality. ComplEx uses complex-valued embeddings specifi-

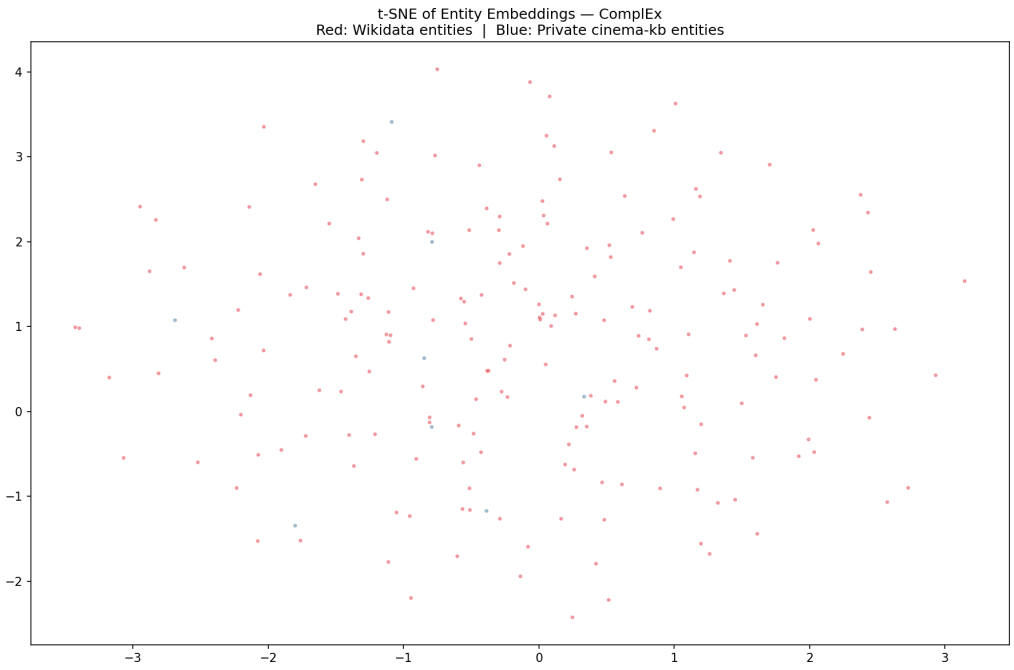

Figure 1: t-SNE projection of TransE entity embeddings. Red: Wikidata entities. Blue: private cinema-kb.org entities.

cally designed for asymmetric and inverse relations, but failed to converge on this dataset due to parameter overload relative to the 220 available training triples. No symmetric or inverse relation pairs were present in the 6-relation vocabulary, which partially explains why TransE's simpler parameterisation was sufficient here.

Critical Reflection

Impact of predicate alignment. Aligning only 5 private predicates to Wikidata using `owl:equivalentProperty` was correct but insufficient to drive a rich expansion. The keyword-based SPARQL alignment initially produced incorrect matches (e.g., `:direct` was matched to P633 — Quebec cultural heritage, not P57 — film director), requiring manual correction. The final alignment was accurate but resulted in only 6 retained relations after filtering, far below the 50–200 target in the lab instructions.

Impact of noisy expansion. The 70,000 triples added by predicate-controlled expansion brought the raw count to 99,106, but **99% of these triples were metadata predicates** (e.g., `schema:dateModified`, `wikibase:statements`) that do not encode cinema-domain knowledge. After strict filtering, only 276 triples survived. A better strategy would have been to restrict SPARQL expansion to a whitelist of predicates from the outset, or to use CONSTRUCT queries returning only cinema-domain facts.

Open-world vs. embedding assumptions. KGE models operate under the **closed-world assumption**: unobserved triples are treated as false during negative sampling. The original KB was built under the open-world assumption (absence of a triple does not mean falsity). This mismatch penalises valid but unobserved facts during training and inflates false-negative rates, reducing MRR.

Rule-Based vs. Embedding-Based Reasoning

SWRL Rule for the Cinema KB

The following Horn rule was designed for the cinema domain with two conditions:

$$\texttt{Film}(?f) \;\wedge\; \texttt{director}(?f,\, ?d) \;\wedge\; \texttt{director}(?f,\, ?d2) \;\rightarrow\; \texttt{workedWith}(?d,\, ?d2)$$

This rule infers that two directors who co-directed the same film have a *workedWith* relationship — analogous to the example in the lab instructions where shared parenthood implies brotherhood.

Embedding Analogy

The embedding-based equivalent asks whether:

$$\vec{v}(\texttt{director}) + \vec{v}(\texttt{producer}) \;\approx\; \vec{v}(\texttt{cast_member})$$

In my analysis, the composed vector `director + producer` was most similar to `cast_member` (cosine ≈ 0.12), suggesting a weak but detectable signal that the embedding has partially captured collaborative cinema roles. The low score reflects the dataset's sparsity rather than a fundamental model failure: with more training data, this analogy would likely strengthen.

The key difference between the two reasoning paradigms is expressiveness vs. scalability: SWRL rules produce **exact, interpretable inferences** but must be written manually for each relation type, while KGE models **generalise automatically** to unseen triples but cannot provide logical explanations or hard guarantees.